

An exact blowup time for the $a = \frac{1}{2}$ generalized Constantin–Lax–Majda equation from single-mode initial data

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Abstract

We consider the inviscid generalized Constantin–Lax–Majda (De Gregorio–Okamoto–Sakajo–Wunsch) equation on the circle at advection parameter $a = \frac{1}{2}$, the unique nontrivial parameter admitting exact pole-dynamics solutions [8]. Building on the exact periodic solutions of Silantyev, Lushnikov, Siegel and Ambrose [11], we identify the single-mode initial datum $\omega_0 = -\sin x$ as the boundary point of their solvable family under an L^2 -matched parameterization, and show that along this path the family’s closed-form collapse times converge to π . Combining the family’s exactness with the local well-posedness theory of [10] and a continuous-dependence sandwich (the required Lipschitz-dependence estimate is written out in Appendix A), we obtain that the solution from $\omega_0 = -\sin x$ has maximal time of existence exactly $T^* = \pi$ — equivalently $T^* = \pi/A$ for $A \cos(x - \varphi)$ — together with the exact leading blowup rate $\|\omega\|_\infty \sim \sqrt{3}/(\pi - t)$ for the family limit. High-resolution spectral computations independently confirm $T^* = \pi$ to $|\Delta T| \approx 1.2 \times 10^{-6}$ and the rate constant to 10 parts per million, with the derived correction exponent $2/3$.

1 Setting and main result

On $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ consider, for mean-zero ω ,

$$\omega_t + a u \omega_x = u_x \omega, \quad u_x = H\omega, \quad \hat{u}_k = -\frac{\hat{\omega}_k}{|k|} \ (k \neq 0), \quad \hat{u}_0 = 0, \quad (1)$$

with H the periodic Hilbert transform ($\hat{H}_k = -i \operatorname{sgn} k$) and $a = \frac{1}{2}$. Equation (1) is the gCLM/Okamoto–Sakajo–Wunsch family [10] interpolating between the Constantin–Lax–Majda equation ($a = 0$) [4] and the De Gregorio equation ($a = 1$).

Theorem 1. *Let $a = \frac{1}{2}$ and $\omega_0(x) = -\sin x$, and let ω be the solution of (1) with $\omega(0) = \omega_0$ given by [10, Theorem 3.1] (Theorem A1 below), maximal in the class $C([0, T^*); H^1(S^1)/\mathbb{R})$. Then*

$$T^* = \pi.$$

Moreover $\omega \in C([0, T]; H^s(S^1))$ for every $s \geq 0$ and every $T < \pi$, while $\int_0^\pi \|H\omega(t)\|_\infty dt = \infty$ and $\|\omega_x(t)\|_{L^2}$ is unbounded as $t \rightarrow \pi^-$. By translation, reflection, and the exact scaling $\omega \mapsto A\omega(x, At)$: $T^(A \cos(x - \varphi)) = \pi/A$ for all $A > 0$, $\varphi \in \mathbb{R}$.*

Proposition 1 (rate). *Along the solvable family approaching $-\sin x$ (Section 3), $\|\omega(t)\|_\infty(t_c - t) \rightarrow \sqrt{3}$, with correction $O((t_c - t)^{2/3})$. The direct computation from single-mode data is consistent with $\|\omega(t)\|_\infty(\pi - t) \rightarrow \sqrt{3}$. The rate is exact for the family limit; for $\omega_0 = -\sin x$ itself it is conjectural, supported numerically at the 10^{-5} level (Section 6).*

Remarks. (i) $a = \frac{1}{2}$ and $a = 0$ are the *only* parameters for which pole-superposition solutions exist [8], as noted in [1]. (ii) The collapse in our limit has spatial scale $(t_c - t)^{1/3}$, matching the known inviscid $a = \frac{1}{2}$ self-similar exponent $\alpha = 1/3$ found independently in [8] and [2]; the family's endgame realizes that self-similar solution. For the broader singularity theory of the family (1) see [5, 3, 6]. (iii) The blowup time from an explicit, non-self-similar, single-Fourier-mode datum appears to be new. [11] parameterize by pole data and do not treat single-mode data; their Remark 2 notes that the point $(v_c(0), \omega_{-2,i}(0)) = (1, 0)$ is a singular point of their implicit solution whose limit depends on the direction of approach — our boundary path (Section 3) is one such direction, along which the datum is $-\sin x$ and the collapse time is π — and the companion paper [1] reports inviscid periodic $a = \frac{1}{2}$ collapse times only numerically, for two-mode data.

2 The solvable family, real reduction

With $X = \tan(x/2)$, [11, §3.1] construct solutions $\omega = 2 \operatorname{Re}[\omega_-]$ of (1) (their (45) with $\nu = 0$; their “ $\sigma = 0$ ” dissipation $-\nu\tilde{\omega}$ vanishes at $\nu = 0$) with

$$\omega_-(x, t) = \omega_{-1}(t) \left[(X - iv_c)^{-1} - (-i - iv_c)^{-1} \right] + \omega_{-2}(t) \left[(X - iv_c)^{-2} - (-i - iv_c)^{-2} \right], \quad (2)$$

subject to the log-cancellation constraint $\omega_{-1} = 2iv_c\omega_{-2}/(1 - v_c^2)$ (their (42)). In the real reduction $\omega_{-2} = iw_2$ (w_2 real), $v_c \in (0, 1)$ real, the pole dynamics reduce to (their (57))

$$w_2' = \frac{w_2^2 (1 - 2v_c^2 + 5v_c^4)}{4v_c^2(1 - v_c^2)^2}, \quad v_c' = -\frac{w_2 (1 + v_c^2)}{4v_c(1 - v_c^2)}, \quad (3)$$

with the implicit solution (their (60), $\nu = 0$)

$$F(v_c(t)) = F(v_c(0)) + R_0 t, \quad F(v) := \frac{v(v^2 - 1)}{(v^2 + 1)^2} + \arctan v, \quad R_0 = \frac{2w_2(0)v_0}{(v_0^2 + 1)^2(v_0^2 - 1)}. \quad (4)$$

A computation gives

$$F'(v) = \frac{8v^2}{(1 + v^2)^3} > 0, \quad F(0) = 0, \quad F(1) = \frac{\pi}{4}, \quad F(\infty) = \frac{\pi}{2}, \quad (5)$$

so F is a strictly increasing bijection $(0, \infty) \rightarrow (0, \pi/2)$. Type-A collapse ($v_c \rightarrow 0^+$) occurs when F reaches 0; there $\|\omega(t)\|_{L^2} \rightarrow \infty$ (their Theorem 3.7), hence $\|\omega(t)\|_\infty \rightarrow \infty$ since $\|\omega\|_{L^2} \leq \sqrt{2\pi} \|\omega\|_\infty$ on S^1 . For $w_2(0) > 0$, $0 < v_0 < 1$ (so $R_0 < 0$) the collapse time is closed-form:

$$t_c(v_0, w_2(0)) = \frac{F(v_0)(v_0^2 + 1)^2(1 - v_0^2)}{2w_2(0)v_0}. \quad (6)$$

We also record the second relation of their (58) at $\nu = 0$, which expresses w_2 through v_c along a trajectory:

$$w_2(t) = w_2(0) \cdot \frac{v_0}{v_c(t)} \cdot \frac{1 - v_c(t)^2}{1 - v_0^2} \cdot \left(\frac{v_c(t)^2 + 1}{v_0^2 + 1} \right)^2. \quad (7)$$

3 The boundary limit

Choice of path. Set

$$w_2(0) = \frac{1 - v_0^2}{2}, \quad v_0 \rightarrow 1^-. \quad (8)$$

Then $\omega_{-1}(0) = -v_0$ exactly, and $R_0 = -v_0/(v_0^2 + 1)^2 \rightarrow -1/4$. (The endpoint $(v_0, w_2(0)) = (1, 0)$ is the singular point of the implicit solution discussed in [11, Remark 2]; (8) fixes one direction of approach to it.)

Lemma 1 (data limit). *With (8), the initial datum $\omega^{(v_0)}(\cdot, 0)$ of the family satisfies, for every $s \geq 0$,*

$$\left\| \omega^{(v_0)}(\cdot, 0) - (-\sin x) \right\|_{H^s(S^1)} \leq C_s (1 - v_0), \quad v_0 \in [\tfrac{1}{2}, 1).$$

Proof. In $Z = e^{ix}$: $X - iv = -i[(1+v)Z - (1-v)]/(Z+1)$, so $(X - iv)^{-1} = i(Z+1)/[(1+v)(Z-r)]$ with $r := (1-v)/(1+v) \in (0, 1/3]$ for $v \geq \frac{1}{2}$. Expanding on $|Z| = 1 > r$: $(Z-r)^{-1} = \sum_{n \geq 1} r^{n-1} Z^{-n}$ and $(Z-r)^{-2} = \sum_{n \geq 2} (n-1)r^{n-2} Z^{-n}$. Substituting into (2) with (8) and collecting mode coefficients $c_{-n}(v_0)$:

- $n = 1$: $c_{-1} = -\frac{2iv}{(1+v)^2} - \frac{i(1-v^2)(1+r)}{(1+v)^2} = -\frac{i}{2} + O(1-v_0)$, using $\frac{2v}{(1+v)^2} = \frac{1}{2} - \frac{(1-v)^2}{2(1+v)^2}$ and $1-v^2 = O(1-v)$;
- $n \geq 2$: $|c_{-n}| \leq C[r^{n-1} + (1-v)(n-1)r^{n-2}] \leq C' n(1-v_0)r^{n-2}$, since $r \leq (1-v)$.

At $v_0 = 1$ ($r = 0$) only $c_{-1} = -i/2$ survives: $\omega_- = -(i/2)e^{-ix}$, $\omega = 2\operatorname{Re}[\omega_-] = -\sin x$. For $v_0 < 1$,

$$\begin{aligned} \left\| \omega^{(v_0)}(\cdot, 0) - (-\sin x) \right\|_{H^s}^2 &= 2 \sum_n n^{2s} |c_{-n} - \delta_{n1}(-i/2)|^2 \\ &\leq C(1-v_0)^2 \left[1 + \sum_{n \geq 2} n^{2s+2} \left(\frac{1}{3}\right)^{2(n-2)} \right] \leq C_s^2 (1-v_0)^2. \quad \square \end{aligned}$$

Lemma 2 (collapse time). *With (8), $t_c(v_0) = F(v_0)(v_0^2 + 1)^2/v_0$ and*

$$t_c(v_0) = \pi - (4 + \pi)(1 - v_0) + o(1 - v_0) \rightarrow \pi \quad \text{as } v_0 \rightarrow 1^-.$$

Proof. Insert (8) into (6): $t_c(v_0) = F(v_0)g(v_0)$ with $g(v) = (v^2 + 1)^2/v$, continuous on $(0, 1]$ with $t_c(1) = F(1)g(1) = (\pi/4) \cdot 4 = \pi$ by (5). For the slope: $g'(v) = (v^2 + 1)(3v^2 - 1)/v^2$, so $g'(1) = 4$; $F'(1) = 1$ by (5); hence $t'_c(1) = F'(1)g(1) + F(1)g'(1) = 4 + \pi$. $\square$

Remark (path dependence of the approach slope). Along the exactly L^2 -matched path $w_2(0) = v_0^{3/2}(1 - v_0^2)/\sqrt{2(1 + v_0^2)}$ — same limit datum — the prefactor $h(v) = (v^2 + 1)^{5/2}/(\sqrt{2}v^{5/2})$ has $h'(1) = 0$, so there $t'_c(1) = F'(1)h(1) = 4$ and $t_c = \pi - 4(1 - v_0) + o(1 - v_0)$. Both paths give $t_c \rightarrow \pi$; only the approach slope is path-dependent.

Lemma 3 (uniform bounds off collapse). *For every $\varepsilon \in (0, \pi/2)$ there exist $\bar{v}(\varepsilon) < 1$, $\delta(\varepsilon) > 0$, $C_s(\varepsilon)$ such that for all $v_0 \in [\bar{v}, 1)$: on $[0, \pi - \varepsilon]$,*

$$v_c(t) \geq \delta(\varepsilon), \quad 0 < w_2(t) \leq \frac{1}{2\delta(\varepsilon)}, \quad \left\| \omega^{(v_0)}(\cdot, t) \right\|_{H^s} \leq C_s(\varepsilon),$$

and $\omega^{(v_0)}(\cdot, t)$ is analytic in the strip $|\operatorname{Im} x| < 2 \operatorname{artanh}(\delta(\varepsilon))$.

Proof. By Lemma 2 choose $\bar{v} \in [1/\sqrt{3}, 1)$ with $t_c(v_0) > \pi - \varepsilon/2$ for $v_0 \geq \bar{v}$. By (8), $|R_0| = v_0/(v_0^2 + 1)^2$, which decreases on $[1/\sqrt{3}, 1]$ from 0.325 to 1/4; hence $|R_0| \geq 1/4$ for $v_0 \in [\bar{v}, 1)$. Since $F(v_c(t)) = F(v_0) + R_0 t$ decreases linearly and vanishes at t_c , $F(v_c(t)) = |R_0|(t_c - t) \geq (1/4)(\varepsilon/2) = \varepsilon/8$ on $[0, \pi - \varepsilon]$. Hence $v_c(t) \geq \delta := F^{-1}(\varepsilon/8)$ by (5). Next, from (3), $w'_2 = w_2^2(1 - 2v_c^2 + 5v_c^4)/(4v_c^2(1 - v_c^2)^2) \geq 0$ because the quartic $5v^4 - 2v^2 + 1$ has no real zeros, so w_2 is non-decreasing and stays positive; then $v'_c < 0$ by (3), so $v_c(t) \leq v_0 < 1$ and (7) with (8) gives

$$w_2(t) = \frac{v_0}{2v_c(t)} (1 - v_c(t)^2) \left(\frac{v_c(t)^2 + 1}{v_0^2 + 1} \right)^2 \leq \frac{1}{2v_c(t)} \leq \frac{1}{2\delta}.$$

The coefficients of (2) are therefore bounded uniformly: $\omega_{-2} = iw_2$ with $|\omega_{-2}| \leq 1/(2\delta)$, and $\omega_{-1} = 2iv_c\omega_{-2}/(1 - v_c^2) = -v_0((v_c^2 + 1)/(v_0^2 + 1))^2$, so $|\omega_{-1}| \leq 1$ (the factor $1 - v_c^2$ cancels).

The field (2) has its poles at $X = \pm iv_c$, i.e. at $|\operatorname{Im} x| = 2 \operatorname{artanh}(v_c) \geq 2 \operatorname{artanh} \delta$; the mode expansion of the proof of Lemma 1, with $v = v_c(t) \in [\delta, 1)$, gives $|c_{-n}(t)| \leq C(\delta) n r(\delta)^{n-2}$, $r(\delta) = (1 - \delta)/(1 + \delta) < 1$, uniformly on $[0, \pi - \varepsilon]$ — geometric decay, hence uniform H^s bounds for every s and strip analyticity. $\square$

4 Local theory

Theorem A1 ([10, Theorem 3.1]; local well-posedness, statement as published). *Let $a \in \mathbb{R}$ be given. For all $\omega_0 \in H^1(S^1)/\mathbb{R}$, there exists a $T > 0$ depending only on a and $\|\omega_{0,x}\|_{L^2}$ such that there exists a unique solution $\omega \in C^0([0, T]; H^1(S^1)/\mathbb{R}) \cap C^1([0, T]; L^2(S^1)/\mathbb{R})$ of (1) with $\omega(0) = \omega_0$.*

Their proof is the Kato–Lai theorem (their §5). $H^1(S^1) \subset L^\infty$ with the sharp constant $c_0 = \pi/\sqrt{6}$ they record in their (6): $\|f\|_\infty \leq c_0 \|f_x\|_{L^2}$.

Theorem A2 ([10, Theorem 3.2]; continuation, statement as published). *Suppose $\omega(0) \in H^1(S^1)/\mathbb{R}$, the solution exists on $[0, T)$, and $\int_0^T \|H\omega(t)\|_\infty dt < \infty$. Then the solution extends to $[0, T + \delta]$ for some $\delta > 0$.*

Theorem A2' ([10, Proposition 5.1]; persistence of regularity, statement as published). *Let m be an integer ≥ 2 . If $\omega_0 \in H^m(S^1)/\mathbb{R}$, then $\sup_{0 \leq t \leq T} \|\omega(t)\|_{H^m} < \infty$ as far as the solution ω exists in $C([0, T]; H^1)$.*

Their proof for $m = 2$: $\|\omega_{xx}(t)\|^2 \leq \|\omega_{xx}(0)\|^2 \exp(2C(1 + |a|) \int_0^t \|\omega_x(s)\| ds)$, the other cases “proved similarly”. Together with the H^m version of Theorem A1 — local existence in $C([0, T]; H^m)$ for a time depending only on $\|\omega_0\|_{H^m}$, obtained exactly as in [10, §5] from the Kato–Lai theorem with $V = H^{m+1}$, $W = H^m$, $X = H^{m-1}$ — and uniqueness, Theorem A2' says that the H^1 solution from an H^m datum belongs to $C([0, T_{\max}); H^m)$, where $T_{\max}$ is its maximal time in the H^1 class.

Theorem A3 (Lipschitz dependence). *For integer $s \geq 3$, if $\omega \in C([0, T]; H^1)$ solves (1) with $K := \sup_{[0, T]} \|\omega(t)\|_{H^s} < \infty$, there exist $\eta, C > 0$ depending only on (a, s, K, T) such that every datum $\varpi_0 \in H^{s-1}$ with $\|\varpi_0 - \omega(0)\|_{H^{s-1}} \leq \eta$ yields a solution ϖ on all of $[0, T]$ with $\|\varpi(t) - \omega(t)\|_{H^{s-1}} \leq C \|\varpi_0 - \omega(0)\|_{H^{s-1}}$.*

Theorem A3 is proved in Appendix A.

5 Proof of Theorem 1

The bound $T^* \geq \pi$. Fix $\varepsilon > 0$ and set $s = 5$. By Lemma 3 the family $\{\omega^{(v_0)}\}_{v_0 \geq \bar{v}}$ is bounded in $C([0, \pi - \varepsilon]; H^5)$; since $\partial_t \omega = -a u[\omega] \omega_x + (H\omega) \omega$ maps bounded sets of H^5 into bounded sets of H^4 , it is moreover bounded in $C^1([0, \pi - \varepsilon]; H^4)$, hence equicontinuous with values in H^4 . By Arzelà–Ascoli (with compactness of $H^5 \subset H^4$ on the compact circle) a subsequence converges in $C([0, \pi - \varepsilon]; H^4)$; the limit solves (1) (the nonlinearity is continuous on $H^4 \subset C^1$), lies in $L^\infty([0, \pi - \varepsilon]; H^5)$, and has datum $-\sin x$ by Lemma 1. By uniqueness (Theorem A1) it is the $-\sin x$ solution, which therefore exists on $[0, \pi - \varepsilon]$; as ε was arbitrary, $T^* \geq \pi$, and by Theorem A2' (the datum being smooth) $\omega \in C([0, T]; H^s)$ for every s and every $T < \pi$.

The bound $T^* \leq \pi$. Suppose $T^* > \pi$. Then $\omega \in C([0, \pi]; H^1)$ and, by Theorem A2' with $m = 3$, $K := \sup_{[0, \pi]} \|\omega(t)\|_{H^3} < \infty$. By Theorem A3 ($s = 3$) there is $\eta > 0$ such that every datum ϖ_0 with $\|\varpi_0 - (-\sin x)\|_{H^2} \leq \eta$ yields a solution on $[0, \pi]$ with $\|\varpi(t)\|_{H^2} \leq K + 1$, in particular $\|\varpi(t)\|_\infty \leq c_0(K + 1)$ on $[0, \pi]$. By Lemma 1 the family data enter this ball for v_0 close to 1. The family solution (2) is a classical (real-analytic) solution on $[0, t_c(v_0))$, hence

coincides with the H^1 solution from the same datum by uniqueness (Theorem A1); yet by [11, Theorem 3.7] its L^2 norm — and hence its L^∞ norm — diverges as $t \rightarrow t_c(v_0)^-$, with $t_c(v_0) < \pi$ by Lemma 2 ($v_0 < 1$) — contradicting the bound on $[0, \pi] \supseteq [0, t_c(v_0)]$. Hence $T^* \leq \pi$, so $T^* = \pi$. Since the solution cannot be continued past π , Theorem A2 gives $\int_0^\pi \|H\omega(t)\|_\infty dt = \infty$, and since $\|H\omega\|_\infty \leq c_0\|(H\omega)_x\|_{L^2} = c_0\|\omega_x\|_{L^2}$ (H is an isometry commuting with ∂_x), $\|\omega_x(t)\|_{L^2}$ is unbounded as $t \rightarrow \pi^-$. $\square$

Corollary 1 (scaling). $\omega \mapsto \lambda\omega(x, \lambda t)$ maps solutions to solutions of (1); translations and $x \mapsto -x$ likewise. Hence $T^*(A \cos(x - \varphi)) = \pi/A$.

6 The rate $\sqrt{3}$: exact for the family, conjectural for the limit datum

Near type-A collapse, with (8) and $v := v_c(t) \rightarrow 0$: from (4)–(5), $F(v) = \frac{8}{3}v^3 + O(v^5)$ and $F(v) = |R_0|(t_c - t) \rightarrow (t_c - t)/4$, so $v^3 = \frac{3}{32}(t_c - t)(1 + O((t_c - t)^{2/3}))$. From (7) with (8), $w_2(t) = \frac{1}{8v}(1 + O(v^2))$ as $v_0 \rightarrow 1$. The double-pole term dominates the sup: writing $x = 2v\xi$ near the collapse point, $\max_\xi |\operatorname{Re}[i/(\xi - i)^2]| = 3\sqrt{3}/8$ at $\xi = 1/\sqrt{3}$, giving

$$\|\omega(t)\|_\infty = \frac{3\sqrt{3}}{4} \cdot \frac{w_2}{v^2} (1 + O(v)) = \frac{\sqrt{3}}{t_c - t} (1 + O((t_c - t)^{2/3})).$$

In fact the whole endgame is self-similar with an explicit profile: with $\xi = x/\ell(t)$, $\ell(t) = 2v(t) = 2(3(t_c - t)/32)^{1/3}$,

$$\omega(x, t) = (t_c - t)^{-1} \Phi(\xi) (1 + o(1)), \quad \Phi(\xi) = -\frac{16}{3} \cdot \frac{\xi}{(1 + \xi^2)^2},$$

whose extremum is $|\Phi(1/\sqrt{3})| = \sqrt{3}$ (consistency check) and whose length exponent $1/3$ is the known $a = \frac{1}{2}$ self-similar exponent [8, 2]; Φ itself should coincide (up to normalization and reflection) with the double-pole self-similar profile of those works — we do not claim it as new. What is new here is its realization as the endgame of the boundary path with the specific normalization tied to the $-\sin x$ datum.

For $\omega_0 = -\sin x$ itself the sandwich controls $[0, \pi - \varepsilon]$ only, so the rate does not transfer automatically. The direct measurement, however, is sharp: refined two-phase runs (rate-adaptive stepping, parabolic refinement of the maximum, resolution-clean fit windows with a core of at least 10 grid cells) give

$$\|\omega\|_\infty (\pi - t) = C + c_1(\pi - t)^{2/3} + \dots,$$

with $C = 1.732061 \pm 9.3 \times 10^{-6}$ at $N = 4096$ (and $1.732071 \pm 1.8 \times 10^{-5}$ at $N = 2048$), i.e. $C = \sqrt{3}$ to 10 ppm, with the correction exponent fitted freely as $\beta = 0.699 \pm 0.017$ against the derived $2/3$, and a correction amplitude $c_1 \approx 0.19$ (0.188 at $N = 2048$, 0.190 at $N = 4096$). We therefore state the rate for the limit datum as a conjecture, supported at the 10^{-5} level with the predicted correction structure.

7 Independent numerics

- (i) Windowed Richardson extrapolation of high-resolution dealiased pseudospectral runs ($N \leq 2048$, $dt \geq 1.25 \times 10^{-5}$, with the analyticity strip monitored) from $\cos x$ gives $T = 3.1415939$, $|T - \pi| = 1.2 \times 10^{-6}$, the error monotone under window, dt and N refinement; the same machinery reproduces the exactly known $T(a = 0) = 2$ as 2.0001 with the same fit systematics.
- (ii) The solver is externally anchored on a family solution: the pseudospectral solution of (1) from family data matches the pole reconstruction (2) to 4.7×10^{-14} – 2.1×10^{-12} through $0.9 t_c$

and reproduces (6) to 7.6×10^{-5} ; our transcription of (2)–(4) and (6) is machine-checked (the constructed field is mean-zero and reproduces the L^2 identity of [11] to 1.4×10^{-16} relative; (3) satisfies (4) along integrated trajectories to 3.7×10^{-13}); the sup-error of Lemma 1 is $1.760 \cdot (1 - v_0)$ at $1 - v_0 \in \{10^{-3}, 10^{-5}, 10^{-7}\}$; the static rate formula of Section 6 holds on family fields to 1.2×10^{-4} at $v = 0.01$. (iii) Shadowing: the $-\sin x$ solution tracks the family solutions with distance $\propto (1 - v_0)$, the normalized distance curves collapsing to 5% through $0.75 t_c$ across $v_0 \in \{0.99, 0.999\}$ — the continuous-dependence mechanism of Section 5 observed directly.

8 Discussion

The π originates in $\arctan(\cdot)$ inside the implicit solution of [11] — equivalently in $\int_0^\infty F' = \pi/2$ with $F' = 8v^2/(1 + v^2)^3$ — filtered through the L^2 -matched boundary path with limiting rate $1/4$. The result gives the (apparently first) exact blowup time for gCLM from a plain Fourier-mode datum, at the unique solvable parameter, and an exact rate constant for the approaching family. The boundary point itself is visible in [11] as the singular point of their implicit solution (their Remark 2); what is added here is the identification of the limit datum along an explicit path, the closed-form collapse-time map (6)–(8) for the whole type-A family, and the sandwich that transfers the family’s collapse time to the single-mode datum. All ingredients beyond standard local theory are elementary computations on the construction of [11].

Data and code availability

The scripts, run records and the numerical census that reproduce every quantitative statement in Sections 6 and 7 are archived at <https://github.com/UsernameLoad/clm-threshold-certificates>.

AI statement

This paper was produced with an AI system. The mathematics, the computations and the text were generated by an AI research agent (Claude, Anthropic) working under my direction; I set the objectives and the verification rules, reviewed the results, and take full responsibility for the content. Every number in Sections 6 and 7 is reproducible from the scripts and records archived with the code. No AI system is an author of this paper.

A Proof of Theorem A3

Throughout, all functions are mean-zero on S^1 ; $\|\cdot\|_\sigma := \|\cdot\|_{H^\sigma}$; $s \geq 3$ is an integer and $\sigma := s - 1 \geq 2$, so H^σ is a Banach algebra embedded in C^1 . The velocity map $\hat{u}_k = -\hat{\omega}_k/|k|$ gives, for mean-zero ω ,

$$\|u[\omega]\|_{\sigma+1} \leq \|\omega\|_\sigma, \quad \partial_x u[\omega] = H\omega, \quad \|H\omega\|_\sigma = \|\omega\|_\sigma, \quad (\text{A.1})$$

H being an isometry on mean-zero H^σ . We use two standard inequalities on the circle, with $\Lambda^\sigma := (1 - \partial_x^2)^{\sigma/2}$ ([7]; textbook form in [9, Ch. 3] — stated there on $\mathbb{R}^n$, the proofs adapt verbatim to the torus): the Kato–Ponce commutator estimate

$$\|[\Lambda^\sigma, f]g\|_{L^2} \leq C_\sigma(\|\partial_x f\|_{L^\infty}\|g\|_{\sigma-1} + \|f\|_\sigma\|g\|_{L^\infty}), \quad (\text{KP})$$

and the tame product estimate $\|fg\|_\sigma \leq C_\sigma(\|f\|_{L^\infty}\|g\|_\sigma + \|f\|_\sigma\|g\|_{L^\infty})$. Time derivatives of Sobolev norms below are computed as in [10, §5]; for solutions of the regularity used here the computation is justified by the standard mollification argument, whose details we omit.

Step 0 (H^σ persistence). Let $\varpi_0 \in H^\sigma$ and let ϖ be the H^1 solution of Theorem A1, maximal on $[0, T_{\max})$. By Theorem A2' (and the H^σ local theory quoted with it), $\varpi \in C([0, T_{\max}); H^\sigma)$ — the H^σ norm stays finite on every compact subinterval of the H^1 existence interval.

Step 1 (difference estimate). Let $\omega \in C([0, T]; H^1)$ be the reference solution with $K = \sup_{[0, T]} \|\omega\|_s < \infty$, and ϖ a second solution as in Step 0. The difference $\delta := \varpi - \omega$ satisfies, by bilinearity,

$$\delta_t = -a u[\varpi] \delta_x - a u[\delta] \omega_x + (H\varpi) \delta + (H\delta) \omega, \quad (\text{A.2})$$

a splitting chosen so that every top-order derivative falls on the reference $\omega \in H^s$, never on ϖ . Pair $\Lambda^\sigma(\text{A.2})$ with $\Lambda^\sigma \delta$ in L^2 :

1. *Transport by ϖ :* $\langle \Lambda^\sigma(u[\varpi]\delta_x), \Lambda^\sigma \delta \rangle = \langle [\Lambda^\sigma, u[\varpi]]\delta_x, \Lambda^\sigma \delta \rangle + \langle u[\varpi] \partial_x \Lambda^\sigma \delta, \Lambda^\sigma \delta \rangle$. The second term equals $-\frac{1}{2} \int (H\varpi) |\Lambda^\sigma \delta|^2 \leq \frac{1}{2} \|H\varpi\|_\infty \|\delta\|_\sigma^2$. For the commutator, (KP) with $f = u[\varpi]$, $g = \delta_x$ gives $\|[\Lambda^\sigma, u[\varpi]]\delta_x\|_{L^2} \leq C(\|H\varpi\|_\infty \|\delta_x\|_{\sigma-1} + \|u[\varpi]\|_\sigma \|\delta_x\|_\infty) \leq C\|\varpi\|_\sigma \|\delta\|_\sigma$, using $\|H\varpi\|_\infty \leq C\|\varpi\|_\sigma$, $\|\delta_x\|_{\sigma-1} \leq \|\delta\|_\sigma$, $\|u[\varpi]\|_\sigma \leq \|\varpi\|_\sigma$, and $\|\delta_x\|_\infty \leq C\|\delta\|_\sigma$ ($\sigma \geq 2$). Paired against $\Lambda^\sigma \delta$, the total cost is $C\|\varpi\|_\sigma \|\delta\|_\sigma^2$.
2. *Stretching of the reference by the difference velocity:* $\|u[\delta] \omega_x\|_\sigma \leq C\|u[\delta]\|_\sigma \|\omega_x\|_\sigma \leq C\|\delta\|_{\sigma-1} \|\omega\|_s \leq CK\|\delta\|_\sigma$ (algebra property; all top derivatives on ω).
3. *The term $(H\varpi)\delta$:* $\leq C\|\varpi\|_\sigma \|\delta\|_\sigma$ by the algebra property and (A.1).
4. *The term $(H\delta)\omega$:* $\leq C\|\delta\|_\sigma \|\omega\|_\sigma \leq CK\|\delta\|_\sigma$.

Summing,

$$\frac{d}{dt} \|\delta\|_\sigma^2 \leq C_1(a, s) (K + \|\varpi(t)\|_\sigma) \|\delta\|_\sigma^2. \quad (\text{A.3})$$

Step 2 (bootstrap). Let ϖ be the maximal solution with datum $\varpi_0 \in H^\sigma$, $\|\varpi_0 - \omega(0)\|_\sigma \leq \eta \leq \frac{1}{2}$ (so $\|\varpi_0\|_\sigma \leq K + \frac{1}{2}$), and let $T_{\max}$ be its maximal time in the H^1 class; by Step 0, $\varpi \in C([0, T_{\max}); H^\sigma)$. Set

$$T' := \sup \{ \tau \in [0, \min(T, T_{\max})) : \|\varpi(t)\|_\sigma \leq K + 1 \text{ on } [0, \tau] \} > 0.$$

On $[0, T')$, (A.3) and Gronwall give $\|\delta(t)\|_\sigma \leq \|\delta(0)\|_\sigma e^{C_1(2K+1)T} =: C_2 \|\delta(0)\|_\sigma$. Choose $\eta := \min(\frac{1}{2}, 1/(2C_2))$; then $\|\delta\|_\sigma \leq \frac{1}{2}$, hence $\|\varpi\|_\sigma \leq K + \frac{1}{2}$ on $[0, T')$. If $T' < \min(T, T_{\max})$, continuity of $t \mapsto \|\varpi(t)\|_\sigma$ gives $\|\varpi(T')\|_\sigma \leq K + \frac{1}{2} < K + 1$, so the bound $\|\varpi\|_\sigma \leq K + 1$ persists beyond T' — contradicting the definition of T' . Hence $T' = \min(T, T_{\max})$. If $T_{\max} \leq T$, then $\|H\varpi\|_\infty \leq C\|\varpi\|_\sigma \leq C(K + 1)$ on $[0, T_{\max})$, so $\int_0^{T_{\max}} \|H\varpi\|_\infty dt < \infty$ and Theorem A2 extends ϖ past $T_{\max}$ — a contradiction. Therefore $T_{\max} > T$, $T' = T$, and $\|\varpi(t) - \omega(t)\|_\sigma \leq C_2 \|\varpi_0 - \omega(0)\|_\sigma$ on $[0, T]$. $\square$

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